Introduction to Econometrics
Weeks 5 and 6

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Based on notes from previous instructors

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Introduction to method of moments

- idea behind method of moments (MM) is to derive estimators based on moment conditions for the population
- by taking sample analog of these conditions in the population
- easiest example is to derive an estimator for $\mu = \mathbb{E}(y)$ (assuming it exists)

MM estimator of expectation

- if the expectation exists, we have:

$$\mu = \mathbb{E}(y) \Leftrightarrow \mathbb{E}(y - \mu) = 0$$

- this suggests for as an estimator $\hat{\mu}$ of μ the one solving:

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\mu}) = 0$$

  which is the sample analog of the moment condition $\mathbb{E}(y - \mu) = 0$

- this gives: $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} y_i$, so sample mean is a method of moments estimator for the expectation

(G)MM ESTIMATOR OF EXPECTATION

- suppose now that we assume that y not only has an expectation but is normally distributed
- in that case as the normal is symmetric, we not only have

$$\mathbb{E}\left(y-\mu\right)=0 \text{ but also } \mathbb{E}\left(y-\mu\right)^3=0$$

- even if these conditions are true, it is unlikely that in finite (and even large) samples, the sample analogs:

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i-\hat{\mu}\right)=0 \text{ and } \frac{1}{n}\sum_{i=1}^{n}\left(y_i-\hat{\mu}\right)^3=0$$

  can exactly be satisfied at the same time, as we have just one parameter $\hat{\mu}$ and 2 equations to satisfy

(G)MM ESTIMATOR OF EXPECTATION

- one estimator that makes use of both moment conditions is

$$\hat{\mu} = \arg\min_{\tilde{\mu}} \left(\frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu}) \right)^2 + \left(\frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu})^3 \right)^2$$

- which is a special case (with $\mathbf{W} = \mathbf{I}$) of estimators of the form

$$\hat{\mu} = \arg\min_{\tilde{\mu}} \begin{bmatrix} \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu}) \\ \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu})^3 \end{bmatrix}' \mathbf{W} \begin{bmatrix} \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu}) \\ \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{\mu})^3 \end{bmatrix}$$

  with $\mathbf{W}$ (called a weighting matrix) symmetric and positive definite[1]

- this in an example of generalized method of moments in which 2 moment conditions are used to estimate the expectation of y

[1] From multivariate calculus, you should remember that this is a quadratic form. See Simon and Blume chapter 16 if you need to refresh about this.

(G)MM estimator of expectation

- use more conditions than parameters can potentially improve efficiency
- although not in the example here because homoscedasticity so that OLS estimator (which here is the mean) is BLUE
- efficiency gains rely on the choice of $\mathbf{W}$

OLS as an MM estimator

- we derived OLS estimator by minimizing a sum of square residuals. Actually, it can be derived as a method of moment estimator
- starting from the predetermined regressors assumption $\mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) = \mathbf{0}$:

OLS as an MM estimator

- we derived OLS estimator by minimizing a sum of square residuals. Actually, it can be derived as a method of moment estimator
- starting from the predetermined regressors assumption $\mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) = \mathbf{0}$:

$$\mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) = \mathbf{0} \Leftrightarrow \mathbb{E}\left(\mathbf{x}_i y_i - \mathbf{x}_i \mathbf{x}_i' \boldsymbol{\beta}\right) = \mathbf{0}$$

- consider the estimator $\hat{\boldsymbol{\beta}}$ solving the sample analog of the above moment condition:

$$\frac{1}{n} \sum_{i=1}^{n} \left(\mathbf{x}_i y_i - \mathbf{x}_i \mathbf{x}_i' \hat{\boldsymbol{\beta}}\right) = \mathbf{0} \Leftrightarrow \hat{\boldsymbol{\beta}} = \left(\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i'\right)^{-1} \sum_{i=1}^{n} \mathbf{x}_i y_i = \mathbf{b}$$

- so OLS can also be derived as a method of moments estimator

IV as an MM estimator

- IV estimator derived in exercise 5 of tutorial 6 in the case where there is one endogenous regressor and one instrument is also a method of moments estimator
- in what follows, we are going to consider generalized method of moments estimators which are a generalization of this in the case where there can be more instruments than endogenous regressors

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Notations

- y_i: dependent variable
- $\mathbf{z}_i$: L-dimensional vector of regressors (some of which endogenous)
- $\boldsymbol{\delta}$: parameter vector of interest
- $\mathbf{x}_i$: K-dimensional vector of instruments (exogenous regressors and at least 1 instrument for each endogenous regressor)
- $\mathbf{w}_i$: vector of unique and non-constant elements of $\left(y_i, \mathbf{z}_i', \mathbf{x}_i'\right)'$

Important: The vector instruments $\mathbf{x}_i$ includes BOTH the exogenous regressors and the variables (called excluded instruments) used as instruments for the endogenous regressors. Exogenous regressors act as instruments for themselves.

Assumptions

- **3.1: linearity**:

$$y_i = \underset{1\times L}{\mathbf{z}_i'} \times \underset{L\times 1}{\boldsymbol{\delta}} + \varepsilon_i$$

- **3.2: random sample**: $\{\mathbf{w}_i\}$ is an i.i.d. stochastic process

- **3.3: instrument exogeneity or instrument validity**: all K variables in $\mathbf{x}_i$ are orthogonal to the error term:

$$\mathbb{E}\left(\underset{K\times 1}{\mathbf{x}_i}\ \varepsilon_i\right) = \mathbf{0}$$
$$\mathbb{E}\left(\mathbf{g}_i\right) = \mathbf{0}$$

where $\mathbf{g}_i \equiv \mathbf{x}_i \varepsilon_i$

- **3.4: rank condition for identification**: the $K \times L$ matrix

$$\boldsymbol{\Sigma}_{xz} = \mathbb{E}\left(\mathbf{x}_i \mathbf{z}_i'\right)$$

is of full column rank (rank equals L, the number of its columns)

- To see why we need this, look at the orthogonality condition:

$$\begin{aligned} \mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) &= \mathbf{0} \\ \underbrace{\mathbb{E}\left(\mathbf{x}_i \mathbf{z}_i'\right)}_{\boldsymbol{\Sigma}_{xz}} \boldsymbol{\delta} &= \underbrace{\mathbb{E}\left(\mathbf{x}_i y_i\right)}_{\boldsymbol{\sigma}_{xy}} \end{aligned}$$

- This is a system of K equations in L unknowns. From linear algebra if a solution exists, this solution is unique if and only if $\boldsymbol{\Sigma}_{xz}$ is of full column rank[2]

- notice that the above equations are assumed to be satisfied for $\boldsymbol{\delta}$ so the rank condition ensures that $\boldsymbol{\delta}$ is the unique solution to this system (i.e. it is identified)

2

Identification

- A necessary condition for identification is the **order condition**:
  - K $\geq$ L
  - # instruments $\geq$ # regressors
  - # excluded instruments $\geq$ # endogenous regressors
  - # orthogonality conditions $\geq$ # parameters
- Depending on whether the order condition is satisfied, the parameters can be
  - **Overidentified**: 3.4 satisfied and $K > L$
  - **Just identified**: 3.4 satisfied and $K = L$
  - **Not identified**: 3.4 NOT satisfied or $K < L$ (either instruments not correlated with endogenous regressors or not enough instruments)

Instrumental variables estimator: $K = L$

- Population moment conditions:

- Call $\hat{\boldsymbol{\delta}}_{IV}$ the estimator solving the sample analogs of these moment conditions:

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i}_{\mathbf{s}_{xy}} - \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{z}_i'}_{\mathbf{S}_{xz}}\hat{\boldsymbol{\delta}}_{IV} = \mathbf{0}$$

$$\mathbf{S}_{xz}\hat{\boldsymbol{\delta}}_{IV} = \mathbf{s}_{xy}$$

- If $K = L$, $\boldsymbol{\Sigma}_{xz}$ is square and invertible and $\mathbf{S}_{xz} \xrightarrow{p} \boldsymbol{\Sigma}_{xz}$

- OLS is just the special case of IV estimator where $\mathbf{x}_i = \mathbf{z}_i$

Final assumption (for asymptotic normality)

Remember, the Lindeberg-Levy CLT requires the existence of a second moment to be applied.

As for OLS, we will end up applying an LLN and a CLT to the sample equivalent of $\mathbb{E}(\mathbf{x}_i\varepsilon_i)$, so the second moment condition we will need is:

- **3.5: finite second moment of orthogonality conditions**: $\mathbf{S} \equiv \mathbb{E}\left(\mathbf{g}_i\mathbf{g}_i'\right) = \mathbb{E}\left(\mathbf{x}_i\varepsilon_i^2\mathbf{x}_i'\right)$ exists and is finite

Under this assumption, the GMM estimator will be asymptotically normal. Hence, we will be able to use it to derive hypothesis tests and confidence intervals.

Why do we focus on *asymptotic* properties of IV estimator? (I)

- For OLS we showed that $\widehat{\boldsymbol{\delta}}_{OLS}$ is unbiased:

$$\begin{aligned}
\mathbb{E}\left(\widehat{\boldsymbol{\delta}}_{OLS}|\mathbf{Z}\right) &= \mathbb{E}\left(\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\mathbf{y}|\mathbf{Z}\right) \\
&= \mathbb{E}\left(\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\left(\mathbf{Z}\boldsymbol{\delta}+\varepsilon\right)|\mathbf{Z}\right) \\
&= \mathbb{E}\left(\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\mathbf{Z}\boldsymbol{\delta}+\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\varepsilon|\mathbf{Z}\right) \\
&= \boldsymbol{\delta}+\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}' \quad \underbrace{\mathbb{E}\left(\varepsilon|\mathbf{Z}\right)}_{0 \text{ by assumption}}
\end{aligned}$$

- Remember: we assumed $\mathbb{E}\left(\varepsilon|\mathbf{Z}\right)=\mathbf{0}$
- Hence, $\mathbb{E}\left(\widehat{\boldsymbol{\delta}}_{OLS}\right)=\mathbb{E}\left(\mathbb{E}\left(\widehat{\boldsymbol{\delta}}_{OLS}|\mathbf{Z}\right)\right)=\boldsymbol{\delta}$

Why do we focus on *asymptotic* properties of IV estimator? (II)

- For IV we cannot obtain similar finite-sample result:

$$\begin{aligned}\mathbb{E}\left(\widehat{\boldsymbol{\delta}}_{IV}|\mathbf{X},\mathbf{Z}\right) &= \mathbb{E}\left(\left(\mathbf{X}'\mathbf{Z}\right)^{-1}\mathbf{X}'\mathbf{y}|\mathbf{X},\mathbf{Z}\right)\\ &= \mathbb{E}\left(\left(\mathbf{X}'\mathbf{Z}\right)^{-1}\mathbf{X}'\left(\mathbf{Z}\boldsymbol{\delta}+\boldsymbol{\varepsilon}\right)|\mathbf{X},\mathbf{Z}\right)\\ &= \boldsymbol{\delta}+\mathbb{E}\left(\left(\mathbf{X}'\mathbf{Z}\right)^{-1}\mathbf{X}'\boldsymbol{\varepsilon}|\mathbf{X},\mathbf{Z}\right)\\ &= \boldsymbol{\delta}+\left(\mathbf{X}'\mathbf{Z}\right)^{-1}\mathbf{X}'\mathbb{E}\left(\boldsymbol{\varepsilon}|\mathbf{X},\mathbf{Z}\right)\end{aligned}$$

- However, $\mathbb{E}\left(\boldsymbol{\varepsilon}|\mathbf{X},\mathbf{Z}\right)$ is *not* assumed equal to zero!
  - Would imply $\mathbb{E}\left(\boldsymbol{\varepsilon}|\mathbf{Z}\right)=\mathbf{0}$
  - IV assumes $\mathbb{E}\left(\mathbf{X}'\boldsymbol{\varepsilon}\right)=\mathbf{0}$ (implied by $\mathbb{E}\left(\boldsymbol{\varepsilon}|\mathbf{X}\right)=\mathbf{0}$)
- Hence, $\mathbb{E}\left(\widehat{\boldsymbol{\delta}}_{IV}\right)=\boldsymbol{\delta}+\mathbb{E}\left(\left(\mathbf{X}'\mathbf{Z}\right)^{-1}\mathbf{X}'\mathbb{E}\left(\boldsymbol{\varepsilon}|\mathbf{X},\mathbf{Z}\right)\right)\neq\boldsymbol{\delta}$

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Generalized Method of Moments

- Unlike for the IV estimator where $K = L$, we are now considering the **overidentified** case where $K > L$, i.e. more instruments than endogenous regressors

- the orthogonality conditions still give the moment conditions

$$\mathbb{E}\left(\mathbf{x}_i y_i\right) - \mathbb{E}\left(\mathbf{x}_i \mathbf{z}_i'\right)\boldsymbol{\delta} = \mathbf{0}$$
$$\boldsymbol{\Sigma}_{xz}\boldsymbol{\delta} = \boldsymbol{\sigma}_{xy}$$

- we are looking for an estimator $\widetilde{\boldsymbol{\delta}}$ that satisfies the sample analogs

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i y_i}_{\mathbf{s}_{xy}} - \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i \mathbf{z}_i'}_{\mathbf{S}_{xz}}\widetilde{\boldsymbol{\delta}} = \mathbf{0}$$
$$\mathbf{S}_{xz}\widetilde{\boldsymbol{\delta}} = \mathbf{s}_{xy}$$

- but in finite (even large) sample, these conditions can usually never be met simultaneously as we have K conditions to be satisfied and $L < K$ parameters

- Rewrite the sample moment conditions as:

$$\underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\left(y_i-\mathbf{z}_i'\widetilde{\boldsymbol{\delta}}\right)}_{\equiv\mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)}=\mathbf{0}$$

- GMM estimator constructed so that the elements of $\mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)$ are as close as possible to $\mathbf{0}$.

- "Closeness" is measured by minimizing a quadratic form, so that **GMM estimator** is:

$$\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)=\underset{\widetilde{\boldsymbol{\delta}}}{\operatorname{argmin}}\; n\times\mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)'\widehat{\mathbf{W}}\mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)$$

  where $\widehat{\mathbf{W}}$ is a weighting matrix that can be random and depend on the sample size (estimated); $\widehat{\mathbf{W}}\xrightarrow{p}\mathbf{W}$, with $\mathbf{W}$ symmetric and positive definite[3]

[3] If $\widehat{\mathbf{W}}$ is diagonal, the GMM estimator minimizes a weighted sum of the square of the elements in $\mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)$.

- First order conditions:

$$\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{s}_{xy} = \mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\widetilde{\boldsymbol{\delta}}$$

  System of L equations in L unknowns

- $\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}$ is nonsingular, because $\mathbf{S}_{xz}$ is of full column rank and $\widehat{\mathbf{W}}$ is symmetric and positive definite

- So system has unique solution that is the GMM estimator $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)$:

$$\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right) = \left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{s}_{xy}$$

- If $K = L$ then $\mathbf{S}_{xz}$ is square and the GMM estimator reduces to the IV estimator

$$\widehat{\boldsymbol{\delta}}_{IV} = \mathbf{S}_{xz}^{-1}\mathbf{s}_{xy}$$

  which does not depend on the weighting matrix

Large sample properties of GMM

- Each different weighting matrix $\widehat{\mathbf{W}}$ yields a different GMM estimator
- We will look at
  - Large sample properties of GMM for given weighting matrix
  - "Optimal" choice of weighting matrix
- We will find that
  - GMM estimators are consistent under assumptions 3.1-3.4 regardless of $\widehat{\mathbf{W}}$
  - $\widehat{\mathbf{W}}$ does affect the standard errors – optimal weighting yields precise estimates

Link between all our estimators

GMM

$$\hat{\underline{\delta}}(\hat{\underline{W}}) = \left(\underline{S}_{xz}' \hat{\underline{W}} \underline{S}_{xz}\right)^{-1} \underline{S}_{xz}' \hat{\underline{W}} \underline{S}_{xy}$$

Note: $\hat{\delta}(\hat{\underline{W}})$ depends on the matrix used to weight the moments (only restriction: must be positive definite)

if use the optimal weighting matrix under heteroskedasticity

Optimal GMM

Same as GMM but using

$$\hat{\underline{W}} = \left(\frac{1}{M} \sum_{i=1}^{M} \hat{\varepsilon}_i^2 \underline{x}_i \underline{x}_i'\right)^{-1}$$

if use the optimal weighting matrix under homoskedasticity

2SLS

Same as (optimal) GMM but weighting matrix simplifies to $\hat{\underline{W}} = \left(\frac{1}{M} \sum_{i=1}^{M} \underline{x}_i \underline{x}_i'\right)^{-1}$

Note: unlike optimal GMM, it can be computed in one step.

if $L = K$

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IV estimator

$$\hat{\underline{\delta}}_{IV} = \underline{S}_{xz}^{-1} \underline{S}_{xy}$$

if $\underline{x} = \underline{z}$

(i.e. if regressors are predetermined)

OLS estimator

$$\underline{b} = \underline{S}_{xx}^{-1} \underline{S}_{xy}$$

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- GMM estimator is given by:

$$\begin{aligned}\widehat{\delta}\left(\widehat{\mathbf{W}}\right) &= \left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{s}_{xy} \\ &= \left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{W}}\frac{1}{n}\left(\sum_{i=1}^{n}\mathbf{x}_i y_i\right)\end{aligned}$$

- How do we prove consistency? [Hint: similar as for OLS]

- Step 1: replace y_i as a function of ε_i to find the expression for the difference between $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)$ and $\boldsymbol{\delta}$.

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- Step 2: Use assumptions and LLN to show that $\bar{\mathbf{g}} \underset{p}{\rightarrow} \mathbf{0}$ and that the other terms also converge in probability

Sketch of step 2

- Assumptions 3.1-3.4 include iid, existence of $\mathbf{\Sigma}_{xz} = \mathbb{E}\left(\mathbf{x}_i \mathbf{z}_i'\right)$ and $\mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) = \mathbf{0}$, so by Kolmogorov's LLN

- also $\widehat{\mathbf{W}} \underset{p}{\rightarrow} \mathbf{W}$
- Using the lemmas from lecture 4, we therefore have:

$$\left(\mathbf{S}_{xz}' \widehat{\mathbf{W}} \mathbf{S}_{xz}\right)^{-1} \mathbf{S}_{xz}' \widehat{\mathbf{W}} \underset{p}{\rightarrow} \left(\mathbf{\Sigma}_{xz}' \mathbf{W} \mathbf{\Sigma}_{xz}\right)^{-1} \mathbf{\Sigma}_{xz}' \mathbf{W}$$

- Finally, using the fact that $\bar{\mathbf{g}} \underset{p}{\rightarrow} \mathbf{0}$, we get:

$$\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right) - \boldsymbol{\delta} \underset{p}{\rightarrow} \mathbf{0}$$

Consistency of GMM estimator

Proposition 1

Under assumptions 3.1-3.4 the GMM estimator is consistent:

$$plim_{n\to\infty}\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right) = \boldsymbol{\delta}$$

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How do we derive asymptotic distribution of GMM? [Hint: same hint]

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- Step 1: replace y_i as a function of ε_i to find the expression for the difference between $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)$ and $\boldsymbol{\delta}$. Done!

How do we derive asymptotic distribution of GMM? [Hint: same hint]

- Step 1: replace y_i as a function of ε_i to find the expression for the difference between $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)$ and $\boldsymbol{\delta}$. Done!

- Step 2: Use assumptions and CLT to show that $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)$ is asymptotically normal and to derive its variance.

Asymptotic normality of GMM estimator

- we have i.i.d., $\mathbb{E}\left(\mathbf{x}_i \varepsilon_i\right) = \mathbf{0}$ and $\mathbf{S} = \mathbb{E}\left(\mathbf{g_i}\mathbf{g}_i'\right) = \mathbb{E}\left(\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2\right)$ exists and is finite (A3.5)
- by Lindeberg-Levy CLT:
- and so, using results used for consistency and lemma 2.4c:[4]

$$\sqrt{n}\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right) - \boldsymbol{\delta}\right) \underset{d}{\rightarrow} \mathcal{N}\left(\mathbf{0}, Avar\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right)\right)$$

with

$$Avar\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right) = \left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}\boldsymbol{\Sigma}_{xz}'\mathbf{W}\mathbf{S}\mathbf{W}\boldsymbol{\Sigma}_{xz}\left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}$$

[4] Reminder: $\mathbf{A}_n \underset{p}{\rightarrow} \mathbf{A}$ and $\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{x} \Rightarrow \mathbf{A}_n\mathbf{x}_n \underset{d}{\rightarrow} \mathbf{A}\mathbf{x}$.

Asymptotic normality of GMM estimator

Proposition 2

Under assumptions 3.1-3.5 the GMM estimator is asymptotically normal:

$$\sqrt{n}\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)-\boldsymbol{\delta}\right)\xrightarrow{d}\mathcal{N}\left(\mathbf{0}, avar\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]\right)$$

where:

- $avar\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]=\left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}\boldsymbol{\Sigma}_{xz}'\mathbf{W}\mathbf{S}\mathbf{W}\boldsymbol{\Sigma}_{xz}\left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}$
- $\boldsymbol{\Sigma}_{xz}=\mathbb{E}\left(\mathbf{x}_i\mathbf{z}_i'\right)$
- $plim\left(\widehat{\mathbf{W}}\right)=\mathbf{W}$
- $\mathbf{S}=\mathbb{E}\left(\mathbf{g}_i\mathbf{g}_i'\right)=\mathbb{E}\left(\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i'\right)$

- As for OLS, we have shown that the estimator is normally distributed with an asymptotic variance which is a function of population-level matrices
- They are not observable, so, in applications, we are going to replace those with matrices computed from our sample who are consistent estimates of these population-level matrices
- As for OLS, one matrix, here $\mathbf{S} = \mathbb{E}\left(\varepsilon_i^2 \mathbf{x}_i \mathbf{x}_i'\right)$, depends on the unobserved error term ε_i. So, we are going to need an additional assumption for $\frac{1}{n}\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i'$ to be a consistent estimator for $\mathbf{S}$

Asymptotic distribution of GMM estimator – consistent estimate of avar $\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]$

$$\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right) - \boldsymbol{\delta} = \left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{W}}\bar{\mathbf{g}}$$

Proposition 3

Suppose we have a consistent estimator $\widehat{\mathbf{S}}$ for $\mathbf{S}$. Then, under random sampling, we can estimate avar $\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]$ consistently by

$$\left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{W}}\widehat{\mathbf{S}}\widehat{\mathbf{W}}\mathbf{S}_{xz}\left(\mathbf{S}_{xz}'\widehat{\mathbf{W}}\mathbf{S}_{xz}\right)^{-1}$$

Consistent estimation of error variance

- For any consistent estimator $\widehat{\boldsymbol{\delta}}$ of $\boldsymbol{\delta}$ define $\widehat{\varepsilon}_i = y_i - z_i'\widehat{\boldsymbol{\delta}}$
- Under 3.1; 3.2 and finite second moment of the regressors ($\mathbb{E}\left(\mathbf{z}_i\mathbf{z}_i'\right)$ is finite) we can show that

Proposition 4

The error variance can be estimated using any consistent estimator $\widehat{\boldsymbol{\delta}}$:

$$\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 \xrightarrow{p} \mathbb{E}\left(\varepsilon_i^2\right)$$

if $\mathbb{E}\left(\varepsilon_i^2\right)$ exists and is finite

Proof: see end of these slides (self-study material)

Estimation of $\mathbf{S}$

- Suppose $\mathbb{E}\left((x_{ik}z_{il})^2\right)$ exists and is finite for all $k = 1, ..., K$ and $l = 1, ..., L$

Proposition 5

Use a consistent estimator $\widehat{\delta}$ to calculate the residual $\widehat{\varepsilon}_i$. Then, under 3.1; 3.2; 3.5 and 3.6:

$$\widehat{\mathbf{S}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\varepsilon}_i^2 \mathbf{x}_i \mathbf{x}_i'$$

is consistent for $\mathbf{S}$

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Optimal weighting matrix $\widehat{\mathbf{W}}$

- results so far have been derived for any weighting matrix $\widehat{\mathbf{W}} \underset{p}{\to} \mathbf{W}$ with $\mathbf{W}$ symmetric and positive definite
- hence, consistency and asymptotic normality hold under A3.1-A3.5 for any symmetric and positive definite weighting matrix
- it does not mean however that all choices of weighting matrix are equivalent
- indeed, choosing $\widehat{\mathbf{W}}$ wisely can reduce the (asymptotic) variance of GMM estimator (i.e. better efficiency)

Optimal weighting matrix $\widehat{\mathbf{W}}$

The asymptotic variances of GMM estimators indexed by $\mathbf{W}$ are given by

$$\text{avar}\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right] = \left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}\boldsymbol{\Sigma}_{xz}'\mathbf{W}\mathbf{S}\mathbf{W}\boldsymbol{\Sigma}_{xz}\left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}$$

Proposition 6

A lower bound on the asymptotic variance is given by $\left(\boldsymbol{\Sigma}_{xz}'\mathbf{W}\boldsymbol{\Sigma}_{xz}\right)^{-1}$*. That bound is achieved if* $\text{plim}\left(\widehat{\mathbf{W}}\right) = \mathbf{S}^{-1}$*:*

$$\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1} = \left(\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2\mathbf{x}_i\mathbf{x}_i'\right)^{-1}$$

Two step estimation procedure

- We need a consistent estimator $\widehat{\mathbf{S}}$ of $\mathbf{S}$ to calculate the efficient GMM estimator
- $\widehat{\mathbf{S}}$ based on any consistent estimator of $\boldsymbol{\delta}$ is consistent for $\mathbf{S}$
- Two step procedure:
  1. Choose any "convenient" matrix $\widehat{\mathbf{W}}$, such as $\widehat{\mathbf{W}} = \mathbf{I}_K$ or $\widehat{\mathbf{W}} = \mathbf{S}_{xx}^{-1}$.[5] Use this $\widehat{\mathbf{W}}$ to compute a consistent estimate $\widehat{\boldsymbol{\delta}}$ of $\boldsymbol{\delta}\left(\widehat{\mathbf{W}}\right)$ and ε_i^2 and $\widehat{\mathbf{S}}$
  2. Use $\widehat{\mathbf{S}}$ to calculate the efficient GMM estimator $\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{S}}^{-1}\right) = \left(\mathbf{S}_{xz}'\widehat{\mathbf{S}}^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\widehat{\mathbf{S}}^{-1}\mathbf{s}_{xy}$

[5] In this case, you are first doing 2SLS (see below). This is the default weighting matrix used by Stata in this first step when you use `ivregress gmm`.

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Hypothesis testing

- the GMM estimator is asymptotically normal
- as OLS was
- as a result, we can derive heteroscedasticity-robust $t-$ and Wald tests exactly in the same way as we did for OLS

Hypothesis testing

$$\sqrt{n}\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)-\boldsymbol{\delta}\right)\overset{a}{\sim}\mathcal{N}\left(\mathbf{0},\text{avar}\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]\right)$$

- $t-$test: under H_0: $\delta_l=\bar{\delta}_l$

$$t_l\equiv\frac{\sqrt{n}\left(\widehat{\delta}\left(\widehat{\mathbf{W}}\right)_l-\bar{\delta}_l\right)}{\sqrt{\widehat{\text{avar}\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]}_{ll}}}=\frac{\widehat{\delta}\left(\widehat{\mathbf{W}}\right)_l-\bar{\delta}_l}{SE_l^*}\xrightarrow{d}\mathcal{N}\left(0,1\right)$$

- Wald test: under H_0: $\mathbf{R}\boldsymbol{\delta}=\mathbf{r}$

$$n\times\left(\mathbf{R}\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)-\mathbf{r}\right)'\left(\mathbf{R}\left(\text{avar}\left[\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)\right]\right)\mathbf{R}'\right)^{-1}\left(\mathbf{R}\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{W}}\right)-\mathbf{r}\right)\xrightarrow{d}\chi^2\left(\#r\right)$$

$\#r$ is the number of restrictions and $\mathbf{R}$ $(\#r\times L)$ is of full row rank

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Hansen (1982) test for overidentifying restrictions

- If parameters are exactly identified ($L = K$) and $\mathbf{\Sigma}_{xz}$ of full column rank the GMM estimator solves orthogonality conditions exactly:

$$J\left(\widetilde{\boldsymbol{\delta}}, \widehat{\mathbf{W}}\right) = n \times \mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right)' \widehat{\mathbf{W}} \mathbf{g}_n\left(\widetilde{\boldsymbol{\delta}}\right) = 0$$

- If # orthogonality conditions $K >$ # parameters L: $J\left(\widetilde{\boldsymbol{\delta}}, \widehat{\mathbf{W}}\right) \neq 0$
- But if orthogonality conditions hold, minimized distance should be *close* to 0
- If we knew $\boldsymbol{\delta}$ and have a consistent estimator for $\mathbf{S}$:

$$J\left(\boldsymbol{\delta}, \widehat{\mathbf{S}}^{-1}\right) = n \times \bar{\mathbf{g}}' \widehat{\mathbf{S}}^{-1} \bar{\mathbf{g}} = \left(\sqrt{n}\bar{\mathbf{g}}\right)' \widehat{\mathbf{S}}^{-1} \left(\sqrt{n}\bar{\mathbf{g}}\right) \xrightarrow{d} \chi^2(K)$$

where $\bar{\mathbf{g}} = \mathbf{g}_n(\boldsymbol{\delta})$. As $\sqrt{n}\bar{\mathbf{g}} \xrightarrow{d} N(0, \mathbf{S})$ and $\widehat{\mathbf{S}} \xrightarrow{p} \mathbf{S}$

Hansen (1982) test for overidentifying restrictions

- Overidentifying restrictions test
  - H_0: $\mathbb{E}(\mathbf{g}_i) = \mathbb{E}(\mathbf{x}_i \varepsilon_i) = \mathbf{0}$
  - Test statistic: for $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$, $J\left(\widehat{\boldsymbol{\delta}}, \widehat{\mathbf{W}}\right) \overset{a}{\sim} \chi^2(K-L)$:

$$J\left(\widehat{\boldsymbol{\delta}}, \widehat{\mathbf{S}}^{-1}\right) = n \times \mathbf{g}_n\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{S}}^{-1}\right)\right)' \widehat{\mathbf{S}}^{-1} \mathbf{g}_n\left(\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{S}}^{-1}\right)\right) \xrightarrow{d} \chi^2(K-L)$$

  - Large J is evidence against the orthogonality assumptions (if we trust the other assumptions)
- DF drop from K to $K-L$: we estimate L parameters $\boldsymbol{\delta}$ before computing the sample average $\mathbf{g}_n$

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Two stage least squares

- Consider now what happens when we have **assumption 3.7 Conditional homoscedasticity**:

$$\mathbb{E}\left(\varepsilon_i^2|\mathbf{x}_i\right) = \sigma_\varepsilon^2$$
$$\mathbf{S} = \mathbb{E}\left(\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i'\right) = \sigma_\varepsilon^2\boldsymbol{\Sigma}_{xx}$$

- Efficient GMM becomes **two stage least squares** (2SLS)

$$\begin{aligned}\widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{S}}^{-1}\right) &= \left(\mathbf{S}_{xz}'\left(\widehat{\sigma}_\varepsilon^2\mathbf{S}_{xx}\right)^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\left(\widehat{\sigma}_\varepsilon^2\mathbf{S}_{xx}\right)^{-1}\mathbf{s}_{xy}\\ &= \left(\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{s}_{xy}\\ &= \widehat{\boldsymbol{\delta}}\left(\widehat{\mathbf{S}}_{xx}^{-1}\right) \equiv \widehat{\boldsymbol{\delta}}_{2SLS}\end{aligned}$$

- $\widehat{\boldsymbol{\delta}}_{2SLS}$ doesn't depend on $\widehat{\sigma}_\varepsilon^2$: can be computed in 1 step

Asymptotic variance of 2SLS estimator (I)

$$\begin{aligned}
\widehat{\boldsymbol{\delta}}_{2SLS} &= \left(\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{s}_{xy} \\
&= \boldsymbol{\delta} + \left(\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\varepsilon_i \\
\widehat{\boldsymbol{\delta}}_{2SLS} - \boldsymbol{\delta} &= \left(\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\mathbf{S}_{xz}\right)^{-1}\mathbf{S}_{xz}'\left(\mathbf{S}_{xx}\right)^{-1}\bar{\mathbf{g}}
\end{aligned}$$

- We derive the asymptotic distribution of $\sqrt{n}\left(\widehat{\boldsymbol{\delta}}_{2SLS} - \boldsymbol{\delta}\right)$ from asymptotic distribution of $\sqrt{n}\bar{\mathbf{g}}$:

$$\begin{aligned}
\sqrt{n}\bar{\mathbf{g}} &\overset{a}{\sim} \mathcal{N}\left(\mathbf{0}, \sigma_\varepsilon^2\boldsymbol{\Sigma}_{xx}\right) \\
\sqrt{n}\left(\widehat{\boldsymbol{\delta}}_{2SLS} - \boldsymbol{\delta}\right) &\overset{a}{\sim} \mathcal{N}\left(\mathbf{0}, \mathbf{A}\sigma_\varepsilon^2\boldsymbol{\Sigma}_{xx}\mathbf{A}'\right)
\end{aligned}$$

where $\mathbf{A} = \left(\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}\boldsymbol{\Sigma}_{xz}\right)^{-1}\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}$

Asymptotic variance of 2SLS estimator (II)

$$\sqrt{n}\left(\widehat{\boldsymbol{\delta}}_{2SLS}-\boldsymbol{\delta}\right)\overset{a}{\sim}\mathcal{N}\left(\mathbf{0},\mathbf{A}\sigma_{\varepsilon}^{2}\boldsymbol{\Sigma}_{xx}\mathbf{A}'\right)$$

where $\mathbf{A}=\left(\underbrace{\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}\boldsymbol{\Sigma}_{xz}}_{\mathbf{B}}\right)^{-1}\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}$

- Notice how avar $\left[\widehat{\boldsymbol{\delta}}_{2SLS}\right]$ simplifies:

$$\begin{aligned}\mathbf{A}\sigma_{\varepsilon}^{2}\boldsymbol{\Sigma}_{xx}\mathbf{A}' &= (\mathbf{B})^{-1}\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}\left(\sigma_{\varepsilon}^{2}\boldsymbol{\Sigma}_{xx}\right)\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}\boldsymbol{\Sigma}_{xz}\left(\mathbf{B}\right)^{-1}\\ &= \sigma_{\varepsilon}^{2}\left(\mathbf{B}\right)^{-1}\mathbf{B}\left(\mathbf{B}\right)^{-1}\\ &= \sigma_{\varepsilon}^{2}\left(\boldsymbol{\Sigma}_{xz}'\left(\boldsymbol{\Sigma}_{xx}\right)^{-1}\boldsymbol{\Sigma}_{xz}\right)^{-1}\end{aligned}$$

- Estimator for asymptotic variance: $\widehat{\text{avar}}\left[\widehat{\boldsymbol{\delta}}_{2SLS}\right]=\widehat{\sigma}_{\varepsilon}^{2}\left(\mathbf{S}_{xz}'\mathbf{S}_{xx}^{-1}\mathbf{S}_{xz}\right)^{-1}$ with $\widehat{\sigma}_{\varepsilon}^{2}=\frac{1}{N-L}\sum_{i=1}^{n}\left(y_i-\mathbf{z}_i'\widehat{\boldsymbol{\delta}}_{2SLS}\right)^{2}$

Sargan's Statistic

Hansen J statistic becomes Sargan's Statistic

$$n \times \frac{\left(\mathbf{s}_{xy} - \mathbf{S}_{xz}\widehat{\boldsymbol{\delta}}_{2SLS}\right)' \mathbf{S}_{xx}^{-1}\left(\mathbf{s}_{xy} - \mathbf{S}_{xz}\widehat{\boldsymbol{\delta}}_{2SLS}\right)}{\widehat{\sigma}_{\varepsilon}^{2}} \xrightarrow{d} \chi^2\left(K - L\right)$$

Computation of Sargan statistic:

1. Estimate the equation $y_i = \mathbf{z}_i'\boldsymbol{\delta} + \varepsilon_i$ by means of 2SLS (with $\mathbf{x}_i$ as instruments). Compute the residuals $\widehat{\varepsilon}_i = y_i - \mathbf{z}_i'\widehat{\boldsymbol{\delta}}_{2SLS}$.

2. Regress $\widehat{\varepsilon}_i$ on $\mathbf{x}_i$ and compute the Sargan statistic as

$$n \times R^2$$

Why call it two stage least squares?

- $\widehat{\boldsymbol{\delta}}_{2SLS}$ can be computed in 2 steps:

  1. **First stage regression**: regress (endogenous) regressor $\mathbf{z}_i$ on the instrument(s) and exogenous regressors $\mathbf{x}_i$. Calculate fitted values $\widehat{\mathbf{z}}_i$

  2. **Second stage regression**: regress the dependent variable y_i on $\widehat{\mathbf{z}}_i$

- remember previous lecture and tutorial where we have seen 2SLS with one regressor (endogenous) and one instrument

- here generalization to multiple regressors and multiple instruments

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Weak instruments

- Check instrument relevance assumption by F-test on **exclusion restrictions** in first stage regression
  - Test joint significance of instruments in the regression explaining the endogenous regressor
  - F statistic should exceed 10
- In general: always important to check instrument relevance
  - GMM is very inefficient if instruments are weakly correlated with endogenous regressors
  - Bias of GMM estimators is large if correlation is very weak
  - See Stock and Yogo (2002) for test statistics

- when we have weak correlation between x and z is that IV estimates will have large standard errors - if correlation is low then there is very little information to estimate the coefficient

- Recall the formula for the $Var(\hat{\beta}_1)$ in the simple regression model. The estimated variance is then:

$$\widehat{Var(\hat{\beta}_1|x,z)} = \frac{\hat{\sigma}^2}{SST_x R_{xz}^2}$$

- In case of no correlation (instrument doesn't satisfy the relevance property) the regression can produce strange results

- On the other hand we don't want the instrument to be too highly correlated with the endogenous variable as then it is almost sure that the exogeneity condition is not satisfied

- Therefore, it is crucial to test relevance every time IV is conducted

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Hausman test for 2SLS

- although consistent, 2SLS is less efficient than OLS if all regressors are exogenous. Hence, OLS preferable over 2SLS if regressors are exogenous

- Hausman test: compare $\widehat{\boldsymbol{\delta}}_{OLS}$ and $\widehat{\boldsymbol{\delta}}_{2SLS}$:

Table. Logic of the Hausman test

	H_0: $\mathbb{E}\left(\mathbf{z}_i\varepsilon_i\right) = \mathbf{0}$	H_a: $\mathbb{E}\left(\mathbf{z}_i\varepsilon_i\right) \neq \mathbf{0}$ (and $\mathbb{E}\left(\mathbf{x}_i\varepsilon_i\right) = \mathbf{0}$)
$\widehat{\boldsymbol{\delta}}_{OLS}$	consistent	inconsistent
$\widehat{\boldsymbol{\delta}}_{2SLS}$	consistent	consistent
	$\text{plim}\left(\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right) = \mathbf{0}$	$\text{plim}\left(\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right) \neq \mathbf{0}$

- Test whether OLS and 2SLS estimates are significantly different
  - If not different: regressors are exogenous – use OLS
  - If different: regressors are endogenous – use 2SLS

Hausman test for 2SLS

- Reject H_0 if the difference $\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}$ is large
- Test statistic (with q is the number of parameters being tested):

$$H = \left(\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right)' \left(\widehat{\mathrm{V}}\left[\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right]\right)^{-1} \left(\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right) \overset{a}{\sim} \chi^2(q)$$

- Tricky bit: estimating $\mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right] = \mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{2SLS}\right] + \mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{OLS}\right] - 2 \times \mathrm{Cov}\left[\widehat{\boldsymbol{\delta}}_{2SLS}, \widehat{\boldsymbol{\delta}}_{OLS}\right]$
  - Usual assumption: one of the estimators is efficient ($\widehat{\boldsymbol{\delta}}_{OLS}$ efficient)
    - Then: $\mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right] = \mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{2SLS}\right] - \mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{OLS}\right]$
    - Assumption of efficiency often not met (OLS not efficient under heteroscedasticity)
  - Heteroscedasticity-robust versions of Hausman test
    - Estimate $\mathrm{V}\left[\widehat{\boldsymbol{\delta}}_{2SLS} - \widehat{\boldsymbol{\delta}}_{OLS}\right]$ (for formula, see Cameron and Trivedi, 2005)
    - Bootstrap
    - Use auxiliary regression from efficient case, but perform the "subset of regressors"-test using robust standard errors

Hausman test for 2SLS

- Computation of Hausman test:
  1. Estimate first stage regression by OLS and obtain the residual
  2. Estimate second stage by OLS and include first stage residual as additional regressor
- Test significance of first stage residual (t-test)
  - Use robust SEs in second step
- If there are multiple RHS variables that may be endogenous: include all first stage residuals and do F-test of joint significance
  - Use robust estimate of covariance matrix

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GMM in practice

- Now you know the theory of GMM
- Credibility of any GMM analysis rests on
  - **Relevance** of the instrument(s)
    - Partial correlation between instruments and endogenous regressors
  - **Validity** of the instrument(s)
    - Test overidentifying restrictions (if applicable)
- The most credible instruments rely on variation that is completely outside control of the subjects being studied
  - Actual **randomized experiments** (randomization done with analysis in mind)
  - **Natural experiments** (randomization occurs by accident)

GMM-related Stata commands

Just-identified case (IV estimator): `ivregress 2sls`

Overidentified case:

- 2SLS: `ivregress`, with option `vce(robust)` to get heteroscedasticity-robust standard errors instead of default non-robust ones
- optimal GMM: `ivregress gmm`

Hausman test:

- non-robust (OLS efficient): `hausman`
- robust (OLS inefficient): `estat endogenous`

First stage estimates:

- `estat firststage` after `ivregress`
- or simply run the fistage regression(s) of endogenous regressor on all instruments

Brief note on IV interpretation with heterogeneity

- so far, we have assumed homogenous effect of z on y, but what if effects are heterogeneous?
- OLS will capture the overall average effect of z on y (if OLS picks up a causal relationship)
- IV will capture the average effect of z on y for those for whom the instrument changes z ; often referred to as a local average treatment effect (LATE)
- if effect of z on y larger (in asbolute value) for this group than for rest of population, IV-type estimator will generally produce larger estimates than OLS estimator
- question is whether this LATE is of interest for the application considered

Readings

- Hayashi, chapter 3: 3.3 (except page 203), 3.4, 3.5 (except (page 214-215), 3.6 (only pages 217-218), 3.8 (except pages 232-234), 3.9

Proof consistent estimation of error variance (I)

The error variance can be estimated using any consistent estimator $\widehat{\boldsymbol{\delta}}$:

$$\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 \xrightarrow{p} \mathbb{E}\left(\varepsilon_i^2\right)$$

Proof:

$$\begin{aligned}\widehat{\varepsilon}_i \equiv y_i - \mathbf{z}_i'\widehat{\boldsymbol{\delta}} &= \mathbf{z}_i'\boldsymbol{\delta} + \varepsilon_i - \mathbf{z}_i'\widehat{\boldsymbol{\delta}} \\ &= \varepsilon_i - \mathbf{z}_i'\left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)\end{aligned}$$

so

$$\widehat{\varepsilon}_i^2 = \varepsilon_i^2 - 2\left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)'\mathbf{z}_i\varepsilon_i + \left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)'\mathbf{z}_i\mathbf{z}_i'\left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)$$

Take sample means

$$\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 = \frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2 - 2\left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)'\frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\varepsilon_i + \left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)'\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\mathbf{z}_i'\right)\left(\widehat{\boldsymbol{\delta}} - \boldsymbol{\delta}\right)$$

Proof consistent estimation of error variance (II)

Apply LLN (evaluate probability limit for $n \to \infty$):

$$\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 = \underbrace{\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2}_{\xrightarrow{p}\mathbb{E}\left(\varepsilon_i^2\right)} - 2\underbrace{\left(\widehat{\boldsymbol{\delta}}-\boldsymbol{\delta}\right)'}_{\xrightarrow{p}\mathbf{0}}\underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\varepsilon_i}_{\xrightarrow{p}\mathbb{E}(\mathbf{z}_i\varepsilon_i)} + \underbrace{\left(\widehat{\boldsymbol{\delta}}-\boldsymbol{\delta}\right)'}_{\xrightarrow{p}\mathbf{0}}\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{z}_i\mathbf{z}_i'\right)}_{\xrightarrow{p}\mathbb{E}\left(\mathbf{z}_i\mathbf{z}_i'\right)}\underbrace{\left(\widehat{\boldsymbol{\delta}}-\boldsymbol{\delta}\right)}_{\xrightarrow{p}\mathbf{0}}$$

so

$$\text{plim}\frac{1}{n}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 = \mathbb{E}\left(\varepsilon_i^2\right)$$

- $\text{plim}\left(\widehat{\boldsymbol{\delta}}-\boldsymbol{\delta}\right) = \mathbf{0}$ since $\widehat{\boldsymbol{\delta}}$ is consistent
- $\mathbb{E}\left(\mathbf{z}_i\varepsilon_i\right)$ and $\mathbb{E}\left(\mathbf{z}_i\mathbf{z}_i'\right)$ are finite under assumptions stated so far